Document Identifier: **PMC8144469**

SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IRB: 2 Methods The study was approved by the institutional Research Subjects Review Board and all subjects signed a written consent prior to undergoing study procedures.
Consent: 2 Methods The study was approved by the institutional Research Subjects Review Board and all subjects signed a written consent prior to undergoing study procedures.
Inclusion and Exclusion Criteria
Participants with severe premorbid or comorbid psychiatric disorders such as diagnoses of schizophrenia, bipolar disorder and active depression were excluded.
Attrition
not detected.
Sex as a biological variable
A total of 122 subjects, 50 recently diagnosed and treatment-naive HIV+ (46 male, mean age 35.54 +- 12.5) and 72 age matched HIV- healthy controls (38 male, mean age 38.89 +- 12) were enrolled.
Subject Demographics
Age: not detected.
Weight: not detected.
Randomization
Linear mixed effects regression (LMER) models were used to quantify the association between selected covariates and continuous response variables, with per-subject random intercepts to account for serial correlation between multiple time points.
Blinding
not detected.
Power Analysis
not detected.

Replication
not required.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
After pre-processing, the group independent component analysis (ICA) of fMRI Toolbox (GIFT, http:mialab.mrn.orgsoftwaregiftindex.html) was used to produce t-score maps reflecting functional maps of 30 components.	GIFT		Suggestion: GIFT (RRID:SCR_024416)(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
The masks were then applied to the pre-processed fMRI images, and the FC was calculated as the Pearson correlation within and between networks.	Pearson correlation
Statistical analysis: Comparisons between two independent groups (e.g., HIV + versus HIV- subjects at baseline) were conducted by either two-group Welchs t-test (for continuous variables such as the concentrations of MRS metabolites) or Fisher exact test (for categorical variables such as sex and race).	Welchs t-test
	Fisher exact test
Paired t-tests were used to compare the levels of continuous variables in HIV + subjects between BSL and W12.	t-tests
In these analyses, parameters were estimated by the restricted maximum likelihood (REML) criterion, and the statistical significance was assessed by the adjusted ANOVA F-test provided by the R package lmerTest (Kuznetsova et al., 2017).	F-test
For inferential problems that involved multiple hypotheses, Benjamini-Hochberg multiple testing procedure (Benjamini and Hochberg, 1995) was used to control the false discovery rate (FDR) at < 0.05 level.	Benjamini-Hochberg
	Benjamini and Hochberg
Association of biomarkers with brain metabolites: We performed Spearman Correlation Test between brain metabolites and average concentrations (pgml) of NfL and Tau at baseline.	Spearman Correlation Test

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq